

Recommendations — cca-mets-scope-unknown-z3p9

Ranked therapeutic options with likelihood, toxicity, mechanism risk, and key references

Generated 2026-06-01

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Profile snapshot

- **Primary site:** biliary tract (cholangiocarcinoma)
- **Histology:** cholangiocarcinoma (adenocarcinoma of the biliary epithelium); intrahepatic vs perihilar vs distal/extrahepatic subsite not specified at intake
- **Stage:** IV (metastatic) per the stated diagnosis; full AJCC TNM staging and metastatic-site inventory not supplied at intake
- **ECOG:** 1
- **Age band:** 60-69
- **Sex:** unknown
- **Biomarkers:** FGFR2 fusion/rearrangement unknown (not yet tested) (ngs_pending); IDH1 R132 mutation unknown (not yet tested) (ngs_pending); HER2 (ERBB2) amplification/overexpression unknown (not yet tested) (ngs_pending); BRAF V600E unknown (not yet tested) (ngs_pending); MSI-H / dMMR unknown (not yet tested) (ngs_pending); Tumor mutational burden (TMB) unknown (not yet tested) (ngs_pending); NTRK1/2/3 fusion unknown (not yet tested) (ngs_pending); RET fusion unknown (not yet tested) (ngs_pending); KRAS G12C unknown (not yet tested) (ngs_pending)
- **Efficacy/toxicity weight:** 0.5

8 ranked therapeutic options across 6 targetable features. Biomarker workup steps live in the standalone Target validation paths report.

Idh1 R132 interventions

1. Ivosidenib (if IDH1 R132 confirmed on tumor NGS) *(conditional on idh1_r132 positive)*

Likelihood of effect: Moderate — only RCT-grade evidence here (ClarIDHy PFS HR 0.37) but the absolute single-agent effect is small (mPFS 2.7 vs 1.4 mo, ORR 2%); benefit is disease control, not shrinkage.

Toxicity burden: Low (QT prolongation ~1% G3, ascites ~7% G3+, fatigue ~3% G3; near-placebo profile)

Counter-productive MoA: Low (No counter-productive vector; dissent is about depth-of-response rank order, not mechanism.)

Overall: The cleanest evidence in the set on a randomized phase 3, but the single-agent payoff is disease stabilization rather than shrinkage; foreclosed if IDH1 R132 is wild-type.

Key references: PMID 32416072 · NCT02989857

Evidence in detail

Ivosidenib (if IDH1 R132 confirmed on tumor NGS)

****Abou-Alfa et al. *Lancet Oncology* 2020****

- **Disease context:**Previously treated advanced cholangiocarcinoma with IDH1 R132 mutation (2L+) — IDH1 R132-mutant CCA after 1-2 prior regimens; crossover from placebo to ivosidenib permitted on progression
- **n:** 185
- **Toxicities:**Ascites (grade ≥ 3): 7.0% (n=121); **Fatigue** (grade ≥ 3): 3.0% (n=121); **Nausea** (grade ≥ 3): 2.0% (n=121); **QT prolongation** (grade 3): 1.0% (n=121); **Serious AE (any)** (grade any): 30.0% (n=121)
- **Efficacy:**HR_PFS 0.37 HR (95% CI 0.25-0.54); **PFS** 2.7 months (95% CI 1.6-4.2); **PFS_rate** 32.0% (95% CI 23.0-42.0); **ORR** 2.0%; **DCR** 53.0%; **HR_OS** 0.69 HR (95% CI 0.44-1.1); **HR_OS** 0.46 HR (95% CI 0.28-0.75)

Fgfr2 Fusion interventions

1. Futibatinib (if FGFR2 fusion/rearrangement confirmed on RNA-based CGP)

(conditional on fgfr2_fusion positive)

Likelihood of effect: High in FGFR2-fusion subset — ORR 42% (FOENIX-CCA2) with cross-trial concordance to pemigatinib (FIGHT-202); the nearest thing to replication in this dossier.

Toxicity burden: Moderate (hyperphosphatemia G3 ~30%, AST increase ~7%, stomatitis/fatigue ~6%)

Counter-productive MoA:Low (Hyperphosphatemia is on-target and manageable, not counter-productive to efficacy; conservative dissent is toxicity-burden, not mechanism.)

Overall:Targets the highest-yield CCA axis with cross-trial-concordant 42% ORR; the on-target 30% grade-3 hyperphosphatemia is the tradeoff against the gentler pemigatinib.

Key references: PMID 36652354 · NCT02052778

2. Pemigatinib (if FGFR2 fusion/rearrangement confirmed) *(conditional on fgfr2_fusion positive)*

Likelihood of effect: High in FGFR2-fusion subset — ORR 35.5% (FIGHT-202), concordant with futibatinib; gentler grade-3 toxicity profile but immature OS.

Toxicity burden: Moderate (hypophosphatemia G3+ ~12%, serous retinal detachment ~4%, PPE/nail toxicity; hyperphosphatemia 55% any-grade with no G3-4)

Counter-productive MoA:Low (On-target FGFR phosphate dysregulation is managed, not counter-productive to the antitumor goal.)

Overall:The same FGFR2-fusion gate as futibatinib with a gentler grade-3 profile and longer track record; the within-lane choice turns on the toxicity-versus-resistance-coverage tradeoff once the fusion confirms.

Key references: PMID 32203698 · NCT02924376

Evidence in detail

Futibatinib (if FGFR2 fusion/rearrangement confirmed on RNA-based CGP)

Goyal et al. *New England Journal of Medicine* 2023

- **Disease context:**Previously treated advanced intrahepatic cholangiocarcinoma with FGFR2 fusion or rearrangement (2L+) — Centrally confirmed FGFR2 fusion/rearrangement in intrahepatic CCA, >=1 prior line; activity retained against some pemigatinib-resistance kinase-domain mutations
- **n:** 103
- **Toxicities:**Hyperphosphatemia (grade >=3): 30.0% (n=103); **Increased AST** (grade >=3): 7.0% (n=103); **Stomatitis** (grade >=3): 6.0% (n=103); **Fatigue** (grade >=3): 6.0% (n=103); **Treatment-related discontinuation** (grade any): 2.0% (n=103)
- **Efficacy:**ORR 42.0% (95% CI 32.0-52.0); **DoR** 9.7 months; **PFS** 9.0 months; **OS** 21.7 months

Pemigatinib (if FGFR2 fusion/rearrangement confirmed)

Abou-Alfa et al. *Lancet Oncology* 2020

- **Disease context:**Previously treated advanced/metastatic cholangiocarcinoma with FGFR2 fusion or rearrangement (2L+) — FGFR2 fusion/rearrangement cohort (cohort A), prior gemcitabine-based chemotherapy; ECOG 0-2
- **n:** 107
- **Toxicities:**Treatment-related grade 3+ AE (any) (grade >=3): 64.0% (n=146); **Hyperphosphatemia** (grade any): 55.0% (n=146); **Hypophosphatemia** (grade >=3): 12.0% (n=146); **Alopecia** (grade any): 46.0% (n=146); **Dysgeusia** (grade any): 38.0% (n=146); **Nail toxicity** (grade any): 42.0% (n=146); **Arthralgia** (grade >=3): 6.0% (n=146); **Stomatitis** (grade >=3): 5.0% (n=146); **Palmar-plantar erythrodysesthesia** (grade any): 11.0% (n=146); **Serous retinal detachment** (grade any): 4.0% (n=146)
- **Efficacy:**ORR 35.5% (95% CI 26.5-45.4); **DCR** 82.0% (95% CI 74.0-89.0); **DoR** 7.5 months (95% CI 5.7-14.5); **PFS** 6.9 months (95% CI 6.2-9.6); **OS** 21.1 months

ErbB2 Amp interventions

1. Zanidatamab (if HER2 IHC 3+ with reflex ISH confirmed) *(conditional on erbb2_amp positive)*

Likelihood of effect: High in HER2-positive subset — confirmed ORR 41.3% with mDoR 12.9 mo in the only dedicated biliary HER2 trial (HERIZON-BTC-01).

Toxicity burden: Low (decreased ejection fraction G3 ~3%, diarrhea G3 ~5%; standard echo monitoring)

Counter-productive MoA:Low (No counter-productive vector; HER2 blockade is direct and on-target with a managed cardiac signal.)

Overall:The guideline-endorsed HER2 lead on dedicated biliary-trial data; gated on HER2 IHC/ISH (not an NGS copy-number call) and on a subsite that drives HER2 prevalence.

Key references: PMID 37423227 · NCT04466891

2. Trastuzumab deruxtecan (if HER2 IHC 3+ confirmed) — kept to document the board's

veto *(conditional on erbb2_amp positive)*

Likelihood of effect: Low-to-moderate as registered (biliary-cohort ORR 22%); the 61.3% IHC 3+ figure is a post-hoc unadjusted subgroup, not the pre-specified endpoint.

Toxicity burden: High (adjudicated ILD/pneumonitis 10.5% with three treatment-related deaths; grade 3+ AEs ~40.8% study-wide)

Counter-productive MoA:High (Conservative veto stands on the uncharacterized ILD long-tail; treatment-related deaths make the toxicity itself the goal-defeating mechanism here.)

Overall:Not recommended as the HER2 lead: a documented ILD signal with three treatment-related deaths and no pulmonary baseline to gate against, where zanidatamab covers HER2 more safely on dedicated biliary data.

Key references: PMID 37870536 · PMID 37423227

Evidence in detail

Zanidatamab (if HER2 IHC 3+ with reflex ISH confirmed)

Harding et al. *Lancet Oncology* 2023

- **Disease context:**Previously treated HER2-amplified biliary tract cancer (2L+) — Cohort 1: HER2 amplified by ISH plus IHC 2+/3+ (4B5 companion antibody); prior gemcitabine-based therapy
- **n:** 80
- **Toxicities:**Diarrhea (grade any): 40.0% (n=80); Diarrhea (grade 3): 5.0% (n=80); Infusion-related reaction (grade any): 35.0% (n=80); Decreased ejection fraction (grade 3): 3.0% (n=80)
- **Efficacy:**ORR 41.3% (95% CI 30.4-52.8); DoR 12.9 months; PFS 5.5 months (95% CI 3.7-7.2)

Trastuzumab deruxtecan (if HER2 IHC 3+ confirmed) — kept to document the board's veto

Meric-Bernstam et al. *Journal of Clinical Oncology* 2024

- **Disease context:**HER2-expressing biliary tract cancer (pan-tumor basket cohort) (2L+) — Biliary tract cohort of a HER2-expressing solid-tumor basket; central HER2 IHC 3+ or 2+. IHC 3+ subgroup responded markedly better than the overall cohort
- **n:** 41
- **Toxicities:**Drug-related grade 3+ AE (any) (grade ≥3): 40.8%; ILD/pneumonitis (grade any): 10.5%; ILD/pneumonitis (grade 5); Nausea (grade any)
- **Efficacy:**ORR 22.0% (95% CI 10.6-37.6); ORR 61.3%; ORR 37.1%; PFS 6.9 months; OS 13.4 months

Harding et al. *Lancet Oncology* 2023

- **Disease context:**Previously treated HER2-amplified biliary tract cancer (2L+) — Cohort 1: HER2 amplified by ISH plus IHC 2+/3+ (4B5 companion antibody); prior gemcitabine-based therapy
- **n:** 80
- **Toxicities:**Diarrhea (grade any): 40.0% (n=80); Diarrhea (grade 3): 5.0% (n=80); Infusion-related reaction (grade any): 35.0% (n=80); Decreased ejection fraction (grade 3): 3.0% (n=80)
- **Efficacy:**ORR 41.3% (95% CI 30.4-52.8); DoR 12.9 months; PFS 5.5 months (95% CI 3.7-7.2)

Braf V600E interventions

1. Dabrafenib + trametinib (if BRAF V600E confirmed on NGS/PCR) *(conditional on braf_v600e positive)*

Likelihood of effect: High in BRAF V600E subset — ORR 47%, mPFS 9.0 mo (ROAR), but on a small single-arm basket cohort (n=43) at 1-5% prevalence.

Toxicity burden: Moderate (pyrexia ~19%, treatment-related serious AEs ~21%, GGT increase G3+ ~12%; any serious AE ~40%)

Counter-productive MoA:Low (No counter-productive vector; the pyrexia/serious-AE load is a patient-AE burden, not a goal-blunting mechanism.)

Overall:Strong 47% ORR on a true V600E with dual guideline-plus-agnostic-label backing; the higher serious-AE and pyrexia load is what keeps it below the gentler oral options at equal agreement.

Key references: PMID 32818466 · NCT02034110

Evidence in detail

Dabrafenib + trametinib (if BRAF V600E confirmed on NGS/PCR)

Subbiah et al. *Lancet Oncology* 2020

- **Disease context:BRAF V600E-mutated biliary tract cancer (2L+)** — Molecularly confirmed BRAF V600E (NGS/PCR; VE1 IHC suggestive but not sufficient); non-V600E class 2/3 BRAF alterations do not qualify
- **n:** 43
- **Toxicities:Serious AE (any)** (grade any): 40.0% (n=43); **Treatment-related serious AE** (grade any): 21.0% (n=43); **Pyrexia** (grade any): 19.0% (n=43); **Increased gamma-glutamyltransferase** (grade ≥3): 12.0% (n=43)
- **Efficacy:ORR** 47.0% (95% CI 31.0-62.0); **DoR** 8.9 months; **PFS** 9.0 months (95% CI 5.0-10.0); **OS** 13.5 months (95% CI 10.0-33.0)

Ntrk Fusion interventions

1. Larotrectinib (if NTRK1/2/3 fusion confirmed on RNA-based CGP) *(conditional on ntrk_fusion positive)*

Likelihood of effect: High in NTRK-fusion subset — ORR 79% pooled, mDoR 35.2 mo; but sub-1% prevalence and no CCA-specific cohort temper the prior.

Toxicity burden: Low (grade 3 AEs ~13%, grade 4 <1%, ALT increase ~3%; no treatment-related deaths)

Counter-productive MoA:Moderate (Critic dissented on cross-tumor extrapolation; acquired NTRK kinase-domain resistance can blunt durability over time.)

Overall:The largest response signal anywhere here if the rare fusion confirms, but the evidence is cross-tumor pooled with no biliary cohort and prevalence is sub-1%.

Key references: PMID 32092306 · NCT02576431

Evidence in detail

Larotrectinib (if NTRK1/2/3 fusion confirmed on RNA-based CGP)

Hong et al. *Lancet Oncology* 2020

- **Disease context:**TRK fusion-positive advanced solid tumors (pooled basket) (2L+) — NTRK1/2/3 fusion across tumor types and ages; biliary patients treated on the fusion irrespective of subsite
- **n:** 159
- **Toxicities:**Grade 3 AE (any) (grade 3): 13.0% (n=260); **Grade 4 AE (any)** (grade 4): 1.0% (n=260); **Increased ALT** (grade ≥3): 3.0% (n=260); **Anemia** (grade ≥3): 2.0% (n=260); **Decreased neutrophils** (grade ≥3): 2.0% (n=260)
- **Efficacy:**ORR 79.0% (95% CI 72.0-85.0); **DoR** 35.2 months; **PFS** 28.3 months; **OS** 44.4 months

Msi Tmb interventions

1. Pembrolizumab (if MSI-H/dMMR or TMB-high confirmed) *(conditional on msi_tmb positive)*

Likelihood of effect: Moderate in MSI-H/TMB-high subset — ORR ~29-34% (KEYNOTE-158) with durable responders, but no CCA-specific subgroup and ~1-3% prevalence.

Toxicity burden: Low (immune-mediated grade 3-5 TRAEs ~14.6%; one treatment-related death)

Counter-productive MoA:Low (No counter-productive vector at this evidence depth; checkpoint resistance is possible but the dissent was preference/evidence-flavored, not mechanistic.)

Overall:Offers a durable immunotherapy tail if MSI-H/dMMR or TMB-high confirms, but the biliary magnitude is inferred from a cross-tumor basket with no CCA-specific subgroup.

Key references: PMID 31682550 · PMID 32919526

Evidence in detail

Pembrolizumab (if MSI-H/dMMR or TMB-high confirmed)

Marabelle et al. *Journal of Clinical Oncology* 2020

- **Disease context:**MSI-H/dMMR advanced noncolorectal solid tumors (incl. biliary) (2L+) — MSI-H/dMMR by PCR or IHC; 27 tumor types, cholangiocarcinoma among the most common. No CCA-specific subgroup ORR published
- **n:** 233
- **Toxicities:**Treatment-related grade 3-5 AE (any) (grade ≥3): 14.6% (n=233); **Treatment-related AE (any)**

(grade any): 64.8% (n=233); **Pneumonia** (grade 5): (n=233)

- **Efficacy:ORR** 34.3% (95% CI 28.3-40.8); **PFS** 4.1 months (95% CI 2.4-4.9); **OS** 23.5 months; **TRAE_rate** 64.8%

****Marabelle et al. *Lancet Oncology* 2020****

- **Disease context:** **TMB-high advanced solid tumors (≥ 10 mut/Mb), incl. biliary (2L+)** — tTMB-high ≥ 10 mut/Mb on FoundationOne CDx; biliary admitted through the biomarker, not a dedicated cohort
- **n:** 102
- **Toxicities:** Safety mirrors the broader KEYNOTE-158 pembrolizumab population; this biomarker analysis did not report separate grade 3+ rates for the TMB-high subgroup.
- **Efficacy:ORR** 29.0% (95% CI 21.0-39.0); **DoR** not reached months; **PFS** 2.1 months (95% CI 2.1-4.1); **OS** 11.7 months (95% CI 9.1-19.1)